Supplementary Appendix

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This appendix has been provided by the authors to give readers additional information about the work.

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Supplementary Appendix

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Criteria for Patient Selection

Inclusion Criteria

Patients had to meet the following criteria at screening:

1. Written informed consent
2. Men aged ≥18 years
3. Histologically or cytologically confirmed adenocarcinoma of prostate
4. Metastatic disease documented either by a positive bone scan or, for soft-tissue or visceral metastases, either by contrast-enhanced abdominal/pelvic/chest computed tomography or magnetic resonance imaging assessed by the investigator and confirmed by central radiology review. Metastatic disease was defined as either malignant lesions in bone scan or measurable lymph nodes above the aortic bifurcation or soft-tissue or visceral lesions according to Response Evaluation Criteria in Solid Tumors v1.1. Lymph nodes were measurable if the short axis diameter was ≥15 mm and soft-tissue/visceral lesions were measurable if the long axis diameter was ≥10 mm.
    a. Patients with regional lymph node metastases only (N1, below the aortic bifurcation) were not eligible for the study. Only patients with non-regional lymph node metastases (M1a) and/or bone metastases (M1b) and/or other sites of metastases with or without bone disease (M1c) were eligible.
5. Patients had to be candidates for androgen-deprivation therapy (ADT) and docetaxel therapy, in the investigator's judgment.
6. Patients had to have started ADT (luteinizing-hormone–releasing hormone [LHRH] agonist or antagonist or orchiectomy) with or without first-generation antiandrogen therapy, but no longer than 12 weeks before randomization. For patients who received LHRH agonists, treatment in combination with a first-generation antiandrogen for at least 4 weeks before randomization was recommended. First-generation antiandrogen therapy had to be stopped before randomization.
7. Eastern Cooperative Oncology Group performance status of 0 or 1
8. Blood counts at screening: hemoglobin ≥9.0 g/dL, absolute neutrophil count ≥1.5 × 10^9/L, platelet count ≥100 × 10^9/L (the patient must not have received any growth factor within 4 weeks or a blood transfusion within 7 days of the hematology laboratory sample obtained at screening)

9. Screening values of serum alanine aminotransferase and/or aspartate transaminase ≤1.5 × the upper limit of normal (ULN), total bilirubin ≤ ULN, and creatinine ≤2.0 × ULN
10. Sexually active men had to have agreed to use condoms as an effective barrier method and refrain from sperm donation, and/or their female partners of reproductive potential had to use a method of effective birth control, during and for 3 months after treatment with darolutamide/placebo and for 6 months after treatment with docetaxel.

Exclusion Criteria

Patients who met any of the following criteria at screening were excluded:

1. Prior treatment with:
    a. LHRH agonist/antagonist started more than 12 weeks before randomization
    b. Second-generation androgen receptor (AR) inhibitors, such as enzalutamide, apalutamide, darolutamide, or other investigational AR inhibitors
    c. CYP17 enzyme inhibitors such as abiraterone acetate or oral ketoconazole as antineoplastic treatment for prostate cancer
    d. Chemotherapy or immunotherapy for prostate cancer before randomization
2. Treatment with radiotherapy (external-beam radiation therapy, brachytherapy, or radiopharmaceuticals) within 2 weeks before randomization
3. Known hypersensitivity to any of the study drugs, study drug classes, or excipients in the formulation of the study drugs
4. Contraindication to both computed tomography and magnetic resonance imaging contrast agent
5. Any of the following within 6 months before randomization: stroke, myocardial infarction, severe or unstable angina pectoris, coronary or peripheral artery bypass graft, or congestive heart failure (New York Heart Association class III or IV)
6. Uncontrolled hypertension, indicated by resting systolic blood pressure ≥160 mmHg or diastolic blood pressure ≥100 mmHg despite medical management
7. Prior malignancy, except for adequately treated basal-cell or squamous-cell carcinoma of the skin or superficial bladder cancer that had not spread behind the connective-tissue layer (i.e., stage pTis, pTa, or pT1) or any cancer for which treatment had been completed ≥5 years before randomization and from which the patient was disease-free

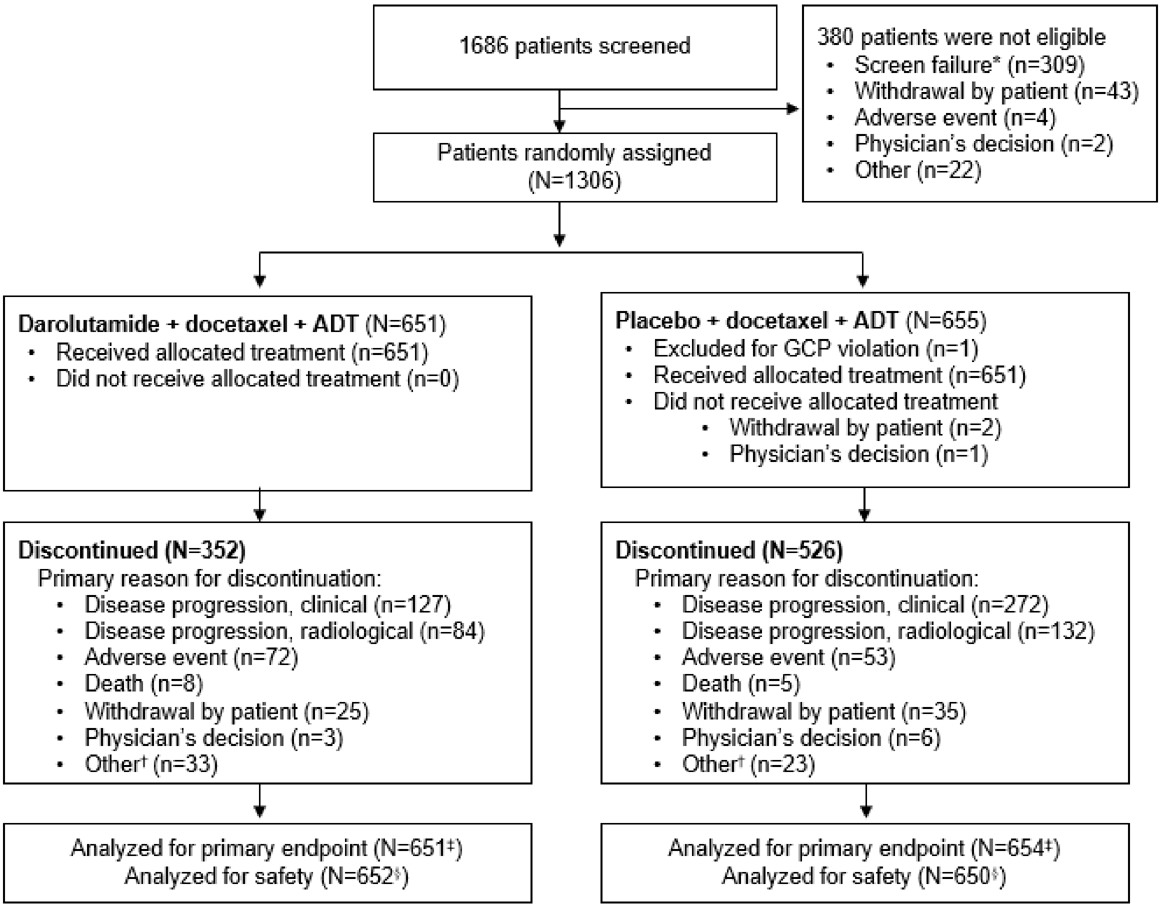
8. A gastrointestinal disorder or procedure that was expected to interfere significantly with absorption of study drug
9. Active viral hepatitis, known human immunodeficiency virus infection with detectable viral load, or chronic liver disease requiring treatment
10. Previous (within 28 days before the start of study drug or 5 half-lives of the investigational treatment of the previous study, whichever was longer) or concomitant participation in another clinical study with investigational medicinal products
11. Any other serious or unstable illness or medical, social, or psychological condition that could jeopardize the safety of the patient and/or their compliance with study procedures or might interfere with their participation in the study or evaluation of the study results
12. Inability to swallow oral medications
13. Close affiliation with the investigational site (e.g., a close relative of the investigator or a dependent person [e.g., employee or student of the investigational site])
14. Previous assignment to treatment in this study

Definitions of Secondary Endpoints

1. **Time to castration-resistant prostate cancer:** the time from randomization to occurrence of the following events, whichever occurred first: prostate-specific antigen progression with serum testosterone at a castrate level (<0.5 ng/mL) or radiological progression of soft-tissue, visceral, or bone lesions; radiological progression by soft tissue/visceral lesions was determined according to RECIST version 1.1 based on magnetic resonance imaging/computed tomography scans of the chest, abdomen, and pelvis performed by the investigator, as recommended by the Prostate Cancer Clinical Trials Working Group (PCWG3); bone lesions were recorded separately from soft-tissue/visceral lesions and determined according to PCWG3 based on whole body ^{99m}Tc methylene diphosphonate bone scans performed by the investigator.
2. **Time to pain progression:** the time from randomization to the first date when a patient experienced pain progression
   a. Pain was assessed with the Brief Pain Inventory–Short Form questionnaire during each visit, before any procedures or examination by the physician.
   b. For asymptomatic patients ("worst pain in 24 hours" score 0 at baseline), pain progression was defined as an increase of ≥2 points in the worst pain score from nadir at two consecutive evaluations ≥4 weeks apart OR initiation of short- or long-acting opioid treatment for cancer pain for ≥7 consecutive days.
   c. For symptomatic patients ("worst pain in 24 hours" score >0 at baseline), pain progression was defined as an increase of ≥2 points in the worst pain score from nadir at two consecutive evaluations ≥4 weeks apart and a worst pain score ≥4 OR initiation of short- or long-acting opioid treatment for cancer pain for ≥7 consecutive days.
3. **Symptomatic skeletal event (SSE)–free survival:** the time from randomization to the first occurrence of an SSE or death from any cause, whichever occurred first
   a. An SSE is defined as external-beam radiation therapy to relieve skeletal symptoms, new symptomatic pathologic bone fracture, or occurrence of spinal cord compression or tumor-related orthopedic surgical intervention, whichever occurred first.
4. **Time to first SSE:** the time from randomization to the first occurrence of an SSE (defined as above)

5. **Time to initiation of subsequent systemic antineoplastic therapy:** the time from randomization to initiation of first subsequent systemic antineoplastic therapy
6. **Time to worsening of disease-related physical symptoms of disease:** the time from randomization to the first date when a patient experienced an increase in disease-related physical symptoms according to the National Comprehensive Cancer Network/Functional Assessment of Cancer Therapy prostate cancer symptom index 17-item questionnaire (NCCN–FACT FPSI–17)
   a. An increase in physical symptoms of disease was defined as a 3-point decrease from baseline in the disease-related physical symptoms subscale (FPSI–DRS–P subscale) at two consecutive evaluations ≥4 weeks apart.
7. **Time to initiation of opioid use for ≥7 consecutive days:** the time from randomization to the start of first opioid use for ≥7 consecutive days
   a. Time to opioid use was determined by opioid use captured via the electronic case report form.
   b. Opioid use for nonmalignant causes was excluded.

Figure S1. CONSORT Flow Diagram



*Patient signed informed consent but did not meet inclusion criteria or met exclusion criteria.

†Other includes noncompliance, additional primary malignancy, lost to follow-up, protocol violation, and other.

‡The primary endpoint (overall survival) was assessed after 533 patients had died (229 in the darolutamide group and 304 in the placebo group). Most deaths were related to disease progression (415/533 [77.9%] in the overall population, 172/229 [75.1%] in the darolutamide group, and 243/304 [79.9%] in the placebo group).

§One patient was randomized in the placebo arm but received darolutamide. The patient was included in the darolutamide group for the safety analysis and the placebo group for the full analysis set.

ADT denotes androgen-deprivation therapy and GCP denotes Good Clinical Practice.

Figure S2. Overall Survival in Prespecified Subgroups (Full Analysis Set, Unstratified)

One patient was randomized in the placebo group, but received darolutamide. The patient was included in the placebo group for the full analysis set. The “rest of the world” geographic region predominantly comprised European countries (Belgium, Bulgaria, Czech Republic, Finland, France, Germany, Italy, Netherlands, Poland, Russian Federation, Spain, Sweden, United Kingdom, Israel, Brazil, Mexico, Australia). ALP denotes alkaline phosphatase; CI denotes confidence interval; ECOG denotes Eastern Cooperative Oncology Group performance status; eCRF denotes electronic case report form; NE denotes not estimable; and PSA denotes prostate-specific antigen.

	Darolutamide+ADT+ Docetaxel	Placebo+ADT+ Docetaxel	Hazard Ratio (95% CI)
	No. of Events/No. of Patients		
Overall	229/651	304/654	0.69 (0.58–0.82)
Extent of disease (eCRF)			
Non-regional lymph node metastasis	5/23	5/16	0.65 (0.19–2.25)
Bone metastasis	171/517	237/520	0.67 (0.55–0.81)
Visceral metastasis	53/111	62/118	0.79 (0.55–1.14)
ALP stratification factor (eCRF)			
Below upper limit of normal	62/290	93/291	0.64 (0.46–0.88)
At or above upper limit of normal	167/361	211/363	0.69 (0.56–0.85)
Age (years)			
<65 yr	80/243	117/234	0.59 (0.45–0.79)
65–74 yr	107/303	124/306	0.82 (0.63–1.06)
75–84 yr	40/102	60/110	0.61 (0.41–0.90)
≥85 yr	2/3	3/4	
Race			
White	131/345	173/333	0.63 (0.50–0.79)
Asian	74/230	93/245	0.84 (0.62–1.14)
Black or African American	8/26	16/28	0.46 (0.20–1.08)
Other or not reported	16/50	22/48	0.58 (0.30–1.11)
Geographical region			
North America	42/125	56/119	0.61 (0.41–0.91)
Asia Pacific	74/229	92/244	0.85 (0.63–1.15)
Rest of the world	113/297	156/291	0.61 (0.48–0.78)
PSA at baseline			
Below median	110/315	142/337	0.77 (0.60–0.98)
At or above median	119/336	162/316	0.62 (0.49–0.78)
ECOG at baseline			
0	150/466	188/462	0.75 (0.61–0.93)
1	79/185	115/190	0.58 (0.43–0.77)
Gleason score			
<8	33/122	44/118	0.65 (0.42–1.03)
≥8	187/505	248/516	0.71 (0.59–0.86)
Metastatic stage at initial diagnosis			
M1 (*de novo*)	206/558	271/566	0.71 (0.59–0.85)
M0 (recurrent)	22/86	30/82	0.61 (0.35–1.05)

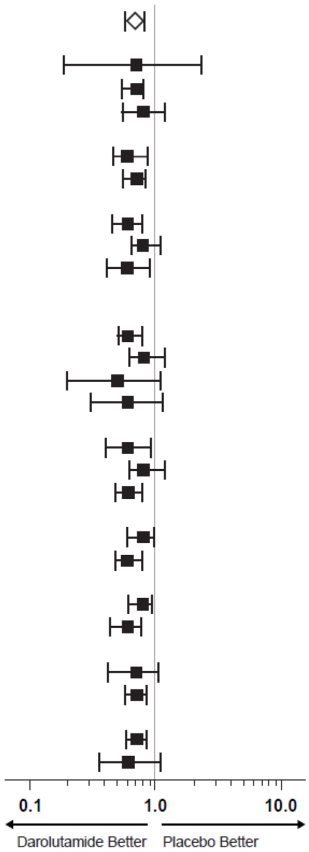
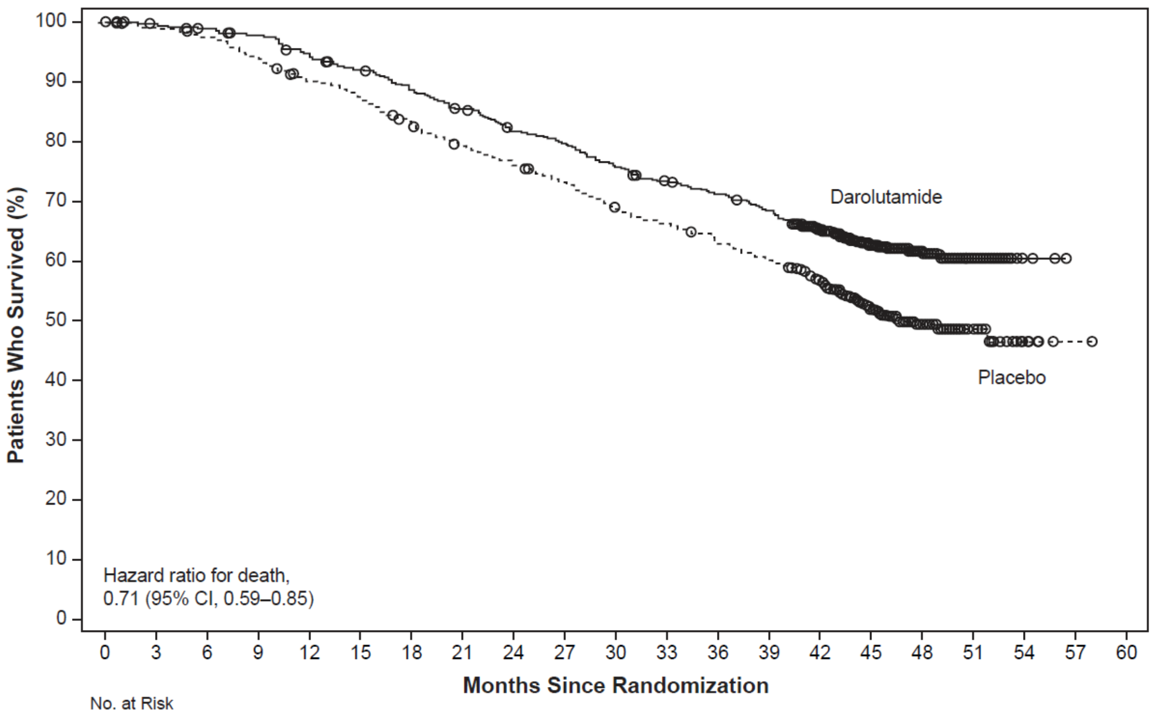


Figure S3. Kaplan–Meier Estimates of Overall Survival by Metastatic Stage at Initial Diagnosis

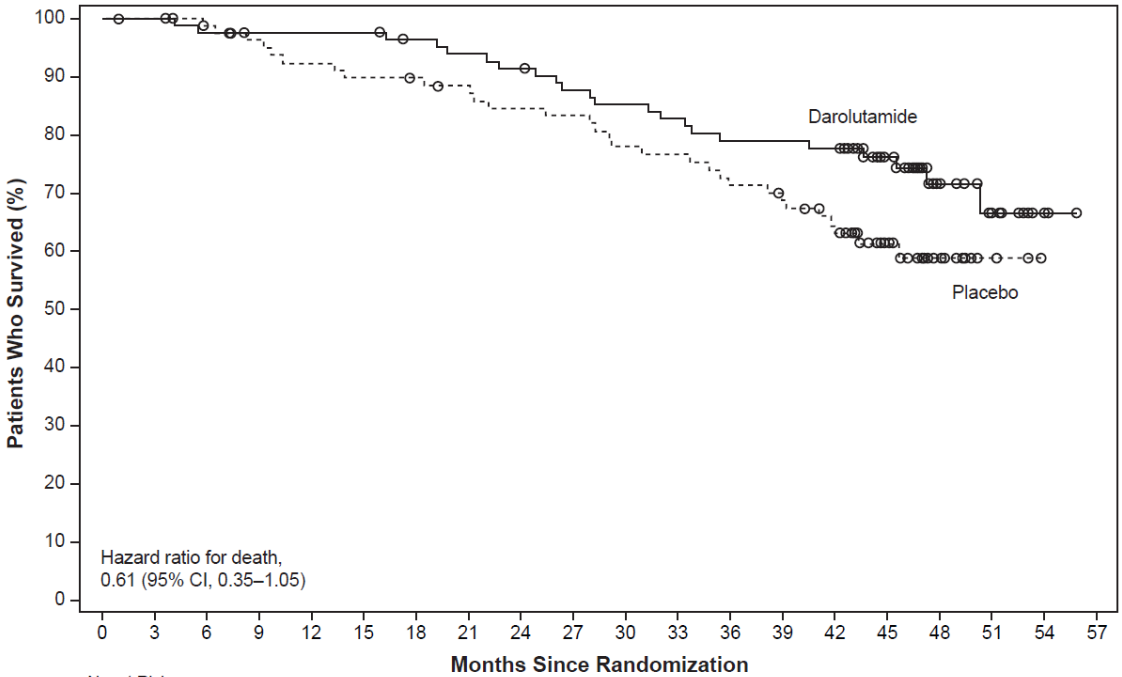
Panel A shows survival in patients with M1 stage (*de novo*) at initial diagnosis. Panel B shows survival in patients with M0 stage (recurrent disease) at initial diagnosis. For missing survival data, patients were censored at their last known date alive. CI denotes confidence interval; mHSPC denotes metastatic hormone-sensitive prostate cancer.

A M1 at initial diagnosis



No. at Risk	0	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54	57	60
Darolutamide	558	553	547	539	520	505	485	466	445	433	412	396	383	367	334	220	116	45	7	0	0
Placebo	566	558	546	526	503	490	461	438	420	403	378	362	344	328	292	190	93	33	6	1	0

B M0 at initial diagnosis



No. at Risk																				
Darolutamide	86	85	83	81	81	81	78	76	74	70	68	66	63	63	62	43	20	11	2	0
Placebo	82	82	78	75	72	70	69	67	64	63	59	58	54	51	45	26	12	4	0	0

Table S1. Participants Randomized by Center

Country	Center Identifier	Total Patients Randomized, *No. of Patients*
Australia	Total	21
	40001	11
	40005	2
	40007	6
	40008	1
	40009	1
Belgium	Total	24
	28001	4
	28002	3
	28005	4
	28006	4
	28007	8
	28008	1
Brazil	Total	53
	50001	3
	50002	11
	50006	4
	50007	9
	50008	1
	50009	9
	50012	4
	50014	3
	50015	9
Bulgaria	Total	17
	41003	1
	41005	8

	41007	1
	41008	2
	41010	5
Canada	Total	26
	26001	8
	26002	8
	26003	4
	26004	4
	26006	2
China	Total	203
	54001	14
	54002	6
	54003	14
	54004	12
	54005	3
	54006	6
	54008	5
	54009	9
	54010	3
	54011	1
	54012	13
	54013	4
	54014	2
	54015	2
	54016	4
	54020	4
	54021	4
	54022	9
	54024	10

	54025	15
	54026	8
	54027	10
	54029	2
	54032	11
	54033	3
	54034	4
	54035	1
	54036	2
	54038	2
	54039	5
	54040	7
	54042	1
	54043	1
	54044	4
	54047	2
Czech Republic	Total	14
	38001	3
	38003	4
	38004	4
	38005	1
	38008	1
	38009	1
Finland	Total	24
	59001	3
	59002	1
	59003	4
	59006	2
	59007	8

	59008	3
	59010	3
France	Total	43
	16001	1
	16002	2
	16004	1
	16005	3
	16006	7
	16009	1
	16011	2
	16012	2
	16013	2
	16015	1
	16016	8
	16017	3
	16018	1
	16019	1
	16021	2
	16023	5
	16024	1
Germany	Total	54
	10001	4
	10002	6
	10003	12
	10005	4
	10007	5
	10009	7
	10010	6
	10011	3

	10013	2
	10014	5
Israel	Total	28
	39001	3
	39002	2
	39003	5
	39004	5
	39005	3
	39006	2
	39008	2
	39009	6
Italy	Total	17
	22001	2
	22002	3
	22003	4
	22005	2
	22006	1
	22010	2
	22011	2
	22012	1
Japan	Total	148
	20001	2
	20002	4
	20003	2
	20004	6
	20005	2
	20006	1
	20007	3
	20008	4

20009	1
20010	1
20011	2
20012	1
20013	4
20014	1
20015	6
20016	4
20018	1
20019	3
20020	5
20021	5
20022	5
20023	15
20024	1
20025	3
20026	3
20027	2
20028	5
20029	3
20030	3
20031	4
20032	2
20033	8
20034	1
20035	1
20036	5
20037	5
20039	1

	20040	7
	20043	4
	20044	3
	20045	1
	20046	4
	20047	2
	20049	2
Mexico	Total	14
	32002	10
	32003	1
	32011	3
Netherlands	Total	33
	30001	2
	30002	6
	30004	7
	30005	6
	30006	2
	30007	2
	30009	3
	30010	5
Poland	Total	17
	18004	1
	18006	4
	18007	1
	18008	2
	18010	2
	18011	7
Russian Federation	Total	91
	51001	19

	51002	28
	51003	6
	51004	7
	51005	4
	51006	3
	51009	2
	51010	8
	51011	10
	51012	4
South Korea	Total	85
	56001	4
	56002	8
	56003	12
	56004	10
	56005	10
	56006	13
	56007	9
	56008	4
	56009	7
	56010	5
	56011	2
	56012	1
Spain	Total	75
	24001	5
	24004	15
	24005	4
	24006	2
	24007	7
	24008	17

	24009	7
	24010	3
	24011	4
	24012	4
	24013	2
	24014	5
Sweden	Total	35
	34001	9
	34002	4
	34003	9
	34004	7
	34005	6
Taiwan	Total	37
	61001	7
	61002	7
	61003	10
	61004	11
	61005	2
United Kingdom	Total	29
	12001	5
	12002	5
	12003	6
	12004	1
	12006	4
	12007	1
	12009	5
	12010	2
United States	Total	218
	14001	3

14004	1
14007	1
14008	2
14013	3
14014	12
14016	7
14017	3
14019	10
14020	4
14021	1
14023	3
14025	2
14026	2
14028	2
14030	3
14031	7
14032	2
14034	1
14035	20
14039	1
14040	2
14041	6
14042	1
14043	2
14044	5
14045	5
14046	2
14047	1

14048	2
14052	5
14054	1
14055	6
14056	4
14057	5
14058	1
14060	1
14061	5
14062	6
14063	4
14064	12
14066	9
14067	6
14068	1
14069	2
14070	1
14071	6
14072	7
14074	1
14076	3
14078	15
14085	1

Table S2. Representativeness of Study Participants

Category	**Description**
Condition under investigation	Metastatic hormone-sensitive prostate cancer that has not yet been treated with androgen-deprivation therapy. This setting may also be referred to as metastatic castrate-sensitive, endocrine-sensitive, or hormone-naive prostate cancer
Special considerations related to:	
Sex and gender	Men (male sex assigned at birth)
Age	Median age at diagnosis of prostate cancer is 67 years
Race or ethnicity	Black and African-American patients have higher incidence rates of prostate cancer and are more likely to die from prostate cancer than patients from other racial and ethnic groups
Geography	Per GLOBOCAN, the global incidence of new cases of prostate cancer in 2020 was estimated at 7.3%. The incidence rates per 100,000 men vary across regions, with the lowest rates in Asian regions (6.3 [South-Central Asia]–28.6% [Western Asia]) and higher rates in North America (73.0%), Australia and New Zealand (75.8%), and Europe (46.4 [Eastern Europe]–83.4% [Northern Europe])
Other considerations	In the US and Europe, 5% of patients have metastatic disease at diagnosis (*de novo*), and 5–15% develop metastatic disease after being treated for localized disease (recurrent); patients with *de novo* metastatic disease account for one-third of prostate cancer-related deaths

Overall representativeness of this study	*Sample size:* 1306 patients – the sample size provided sufficient power to detect a difference between treatment groups *Median age (range):* 67 (41–89) years – consistent with the average age of US patients with prostate cancer *Race:* Asian populations well represented (35%), Black/African Americans well represented among North American participants (22%) *Geography:* 23 countries globally, including North America (26 US states + DC; Canada), Asia-Pacific (China, Japan, South Korea, and Taiwan), and rest of world (13 European countries, Australia, Brazil, Israel, Mexico, but not Africa) – good representation of different global regions and healthcare systems *Metastases at diagnosis:* most patients (86%) had *de novo* metastatic disease; these patients have a poor prognosis and thus an urgent need for effective life-prolonging therapy

DC denotes District of Columbia; GLOBOCAN denotes Global Cancer Incidence, Mortality and Prevalence; and US denotes United States.

Table S3. Subsequent Life-prolonging Systemic Antineoplastic Therapy*

Subsequent Antineoplastic Therapy	Darolutamide + ADT + Docetaxel (N = 315†)	Placebo + ADT + Docetaxel (N = 495†)
No. (%) of patients with subsequent life-prolonging systemic antineoplastic therapy	179 (56.8)	374 (75.6)
Abiraterone acetate	112 (35.6)	232 (46.9)
Enzalutamide	48 (15.2)	136 (27.5)
Cabazitaxel	57 (18.1)	89 (18.0)
Docetaxel	46 (14.6)	89 (18.0)
Radium-223	19 (6.0)	34 (6.9)
Sipuleucel-T	4 (1.3)	10 (2.0)
Lutetium-177 PSMA	2 (0.6)	7 (1.4)
Apalutamide	2 (0.6)	2 (0.4)

*At the data cut-off date (October 25, 2021), 45.9% of patients in the darolutamide group and 19.1% of patients in the placebo group were receiving ongoing study treatment.

†The denominators are the number of patients who entered active or long-term (survival) follow-up, plus one patient who did not enter follow-up but received subsequent therapy. Patients could receive more than one subsequent life-prolonging systemic antineoplastic therapy.

ADT denotes androgen-deprivation therapy and PSMA denotes prostate-specific membrane antigen.

Table S4. Treatment-Emergent Grade 5 Adverse Events (Safety Analysis Set)

Grade 5 Adverse Event	Darolutamide + ADT + Docetaxel (N = 652*)	Placebo + ADT + Docetaxel (N = 650*)
	No. of patients (%)	
Any	27 (4.1)	26 (4.0)
COVID-19 pneumonia	4 (0.6)	1 (0.2)
Sudden death	2 (0.3)	3 (0.5)
General physical health deterioration	1 (0.2)	4 (0.6)
Cardiac arrest	1 (0.2)	2 (0.3)
Death	1 (0.2)	2 (0.3)
Hypoxia	1 (0.2)	1 (0.2)
Acute cardiac failure	1 (0.2)	0
Acute myocardial infarction	1 (0.2)	0
Anesthetic cardiac complication	1 (0.2)	0
Arteriosclerosis	1 (0.2)	0
COVID-19	1 (0.2)	0
Cardiac disorder	1 (0.2)	0
Esophageal carcinoma	1 (0.2)	0
Gastric cancer	1 (0.2)	0
Hemoptysis	1 (0.2)	0
Hemorrhagic stroke	1 (0.2)	0
Hypertension	1 (0.2)	0
Metastases to central nervous system	1 (0.2)	0
Myocardial infarction	1 (0.2)	0
Parkinson's disease	1 (0.2)	0
Pneumonitis	1 (0.2)	0

Respiratory distress	1 (0.2)	0
Septic shock	1 (0.2)	0
Subarachnoid hemorrhage	1 (0.2)	0
Transitional cell carcinoma	1 (0.2)	0
Pneumonia	0	2 (0.3)
Pulmonary sepsis	0	2 (0.3)
Acute kidney injury	0	1 (0.2)
Cachexia	0	1 (0.2)
Cardiac failure	0	1 (0.2)
Cerebrovascular accident	0	1 (0.2)
Dyspnea	0	1 (0.2)
Gastric ulcer perforation	0	1 (0.2)
Hypertransaminasemia	0	1 (0.2)
Interstitial lung disease	0	1 (0.2)
Respiratory failure	0	1 (0.2)
Sepsis	0	1 (0.2)
Urinary bladder hemorrhage	0	1 (0.2)

*Three randomized patients were never treated; all three were in the placebo group. One patient was randomized in the placebo group, but received darolutamide. The patient was included in the darolutamide group for the safety analysis set.

ADT denotes androgen-deprivation therapy.

Table S5. Treatment-Emergent Adverse Events of Any Grade that Occurred in ≥10% of Patients in Either Group, with Exposure-Adjusted Incidence Rates (Safety Analysis Set)

Adverse Event*	Darolutamide + ADT + Docetaxel (N = 652†)		Placebo + ADT + Docetaxel (N = 650†)	
	No. of patients (%)	*EAIR/ 100 PY‡*	*No. of patients (%)*	*EAIR/ 100 PY‡*
Alopecia	264 (40.5)	15.3	264 (40.6)	22.0
Neutropenia§	256 (39.3)	NA¶	252 (38.8)	NA¶
Fatigue	216 (33.1)	12.5	214 (32.9)	17.8
Anemia	181 (27.8)	10.5	163 (25.1)	13.6
Arthralgia	178 (27.3)	10.3	174 (26.8)	14.5
Peripheral edema	173 (26.5)	10.0	169 (26.0)	14.1
Diarrhea	167 (25.6)	9.6	156 (24.0)	13.0
Constipation	147 (22.5)	8.5	130 (20.0)	10.8
Hot flush	124 (19.0)	7.2	122 (18.8)	10.2
Back pain	123 (18.9)	7.1	123 (18.9)	10.2
Decreased appetite	121 (18.6)	7.0	85 (13.1)	7.1
Increased weight	116 (17.8)	6.7	102 (15.7)	8.5
Nausea	115 (17.6)	6.6	133 (20.5)	11.1
Increased alanine aminotransferase	102 (15.6)	5.9	84 (12.9)	7.0
Pain in extremity	98 (15.0)	5.7	78 (12.0)	6.5
Increased aspartate aminotransferase	91 (14.0)	5.3	68 (10.5)	5.7
Pyrexia	86 (13.2)	5.0	90 (13.8)	7.5
Hypertension	85 (13.0)	4.9	59 (9.1)	4.9
Cough	84 (12.9)	4.9	73 (11.2)	6.1
Bone pain	81 (12.4)	4.7	84 (12.9)	7.0

Peripheral neuropathy	76 (11.7)	4.4	67 (10.3)	5.6
Insomnia	74 (11.3)	4.3	81 (12.5)	6.7
Hyperglycemia	74 (11.3)	4.3	61 (9.4)	5.1
Myalgia	73 (11.2)	4.2	63 (9.7)	5.2
Dysgeusia	69 (10.6)	4.0	80 (12.3)	6.7
Asthenia	68 (10.4)	3.9	65 (10.0)	5.4
Stomatitis	66 (10.1)	3.8	57 (8.8)	4.7
Peripheral sensory neuropathy	65 (10.0)	3.8	67 (10.3)	5.6
Urinary tract infection	61 (9.4)	3.5	67 (10.3)	5.6
Dyspnea	59 (9.0)	3.4	71 (10.9)	5.9
Malaise	57 (8.7)	3.3	66 (10.2)	5.5

*Patients may have had more than one treatment-emergent adverse event. Adverse events are described using the Medical Dictionary for Regulatory Activities preferred terms.

†Three randomized patients were never treated; all three were in the placebo group. One patient was randomized in the placebo group, but received darolutamide. The patient was included in the darolutamide group for the safety analysis set.

‡EAIR of treatment-emergent adverse events is defined as the number of patients with a given event, divided by the total darolutamide/placebo treatment duration of all patients in years. The rate is expressed in 100 PY.

§Neutropenia includes preferred terms of leukopenia, neutropenia, decreased neutrophil count, and decreased white blood cell count.

¶Exposure adjustment not applicable, as the majority of adverse events occurred during concomitant docetaxel treatment.

ADT denotes androgen-deprivation therapy; EAIR denotes exposure adjusted incidence rate; NA denotes not applicable; and PY denotes patient years.

Table S6. Adverse Events of Special Interest (Safety Analysis Set)

Adverse Event	Darolutamide + ADT + Docetaxel (N = 652*)		Placebo + ADT + Docetaxel (N = 650*)	
	No. of patients (%)	*EAIR/ 100 PY†*	*No. of patients (%)*	*EAIR/ 100 PY†*
Events commonly associated with ADT or androgen receptor pathway inhibitor therapy				
Fatigue	216 (33.1)	12.5	214 (32.9)	17.8
Vasodilatation and flushing	133 (20.4)	7.7	141 (21.7)	11.7
Rash‡	108 (16.6)	6.2	88 (13.5)	7.3
Diabetes mellitus and hyperglycemia	99 (15.2)	5.7	93 (14.3)	7.7
Hypertension§	89 (13.7)	5.1	60 (9.2)	5.0
Cardiac disorder	71 (10.9)	4.1	76 (11.7)	6.3
Cardiac arrhythmia§	52 (8.0)	3.0	55 (8.5)	4.6
Coronary artery disorder§	19 (2.9)	1.1	13 (2.0)	1.1
Heart failure§	4 (0.6)	0.2	13 (2.0)	1.1
Bone fracture¶	49 (7.5)	2.8	33 (5.1)	2.7
Falls, including accident	43 (6.6)	2.5	30 (4.6)	2.5
Mental-impairment disorder§	23 (3.5)	1.3	15 (2.3)	1.2
Weight decreased	22 (3.4)	1.3	35 (5.4)	2.9
Depressed-mood disorder§	21 (3.2)	1.2	24 (3.7)	2.0
Breast disorders/gynecomastia§	21 (3.2)	1.2	10 (1.5)	0.8
Cerebral ischemia	8 (1.2)	0.5	8 (1.2)	0.7
Seizure	4 (0.6)	0.2	1 (0.2)	0.1

*Three randomized patients were never treated; all three were in the placebo group. One patient was randomized in the placebo group, but received darolutamide. The patient was included in the darolutamide group for the safety analysis set.

[†]EAIR of treatment-emergent adverse events is defined as the number of patients with a given event, divided by the total darolutamide/placebo treatment duration of all patients in years. The rate is expressed in 100 PY.

[‡]This category combines the following MedDRA terms: rash, maculopapular rash, drug eruption, pruritic rash, erythematous rash, macular rash, papular rash, follicular rash, pustular rash, and vesicular rash.

[§]This category is a MedDRA High-Level Group Term.

[¶]Excluding pathologic fractures. This category combines the following MedDRA terms: any fractures and dislocations, limb fractures and dislocations, pelvic fractures and dislocations, skull fractures, facial bone fractures and dislocations, spinal fractures and dislocations, and thoracic cage fractures and dislocations.

ADT denotes androgen-deprivation therapy; EAIR denotes exposure adjusted incidence rate; MedDRA denotes Medical Dictionary for Regulatory Activities; and PY denotes patient years.